

Crop Yield Prediction Using Machine Learning and Climate Data

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Abstract—Accurate prediction of crop yield is essential for improving agricultural planning, ensuring food security, and mitigating the impacts of climate variability. In Bangladesh, where agriculture supports the livelihoods of millions and rice dominates national food production, even small inaccuracies in yield estimation can significantly influence decisions at both farm and policy levels. Despite its importance, many existing machine learning-based studies treat Bangladesh as part of broader regional analyses rather than focusing on it as a dedicated study context. This study proposes a machine learning-based framework for crop yield prediction using historical agricultural and climatic data specific to Bangladesh. The approach involves data preprocessing and exploratory analysis to understand variable distributions and relationships, followed by feature engineering techniques such as interaction features, temporal trend variables, and logarithmic transformations to capture non-linear patterns in the data. Additionally, Principal Component Analysis is explored as a dimensionality reduction method to assess its impact on model performance. Multiple supervised learning algorithms, including Random Forest, AdaBoost, K-Nearest Neighbors (KNN), and Support Vector Machine (SVM), are implemented and evaluated using RMSE and R^2 metrics. The results demonstrate that ensemble methods, particularly Random Forest, achieve superior predictive performance by effectively capturing complex relationships among climatic and agricultural variables, while dimensionality reduction provides limited additional benefit. The findings highlight the importance of feature engineering and model selection in yield prediction tasks and suggest that further improvements can be achieved through the use of more refined and enriched datasets. The proposed framework offers a data-driven approach that can support farmers, researchers, and policymakers in enhancing agricultural productivity and strengthening food security in Bangladesh.

Index Terms—crop yield prediction, Bangladesh, machine learning, SVM, AdaBoost, decision trees, ARIMA, association rule mining, food security, PCA

I. INTRODUCTION

Agriculture remains one of the most critical sectors for global economic stability and food security. As the global population continues to grow, the demand for agricultural products is increasing significantly. Accurate crop yield prediction has therefore become an important research problem in modern agricultural analytics.

In countries such as Bangladesh, agriculture plays a vital role in the national economy and the livelihoods of millions of people. Crop production in this region is strongly affected

by seasonal rainfall patterns, temperature fluctuations, soil fertility, and the increasing impacts of climate change. As a result, traditional statistical approaches often struggle to capture the complex relationships among these variables, making accurate yield prediction challenging. Advances in data science and machine learning have created new opportunities to analyze large agricultural datasets and uncover hidden patterns that influence crop productivity.

Machine learning techniques have shown considerable promise in addressing agricultural prediction problems. Unlike conventional statistical methods, machine learning models can learn non-linear relationships between input variables and output targets, making them suitable for complex real world systems such as crop growth. Algorithms such as Support Vector Machines, decision trees, ensemble methods, and boosting techniques have been widely applied in agricultural research to predict crop yield.

Another important aspect of agricultural analytics is the extraction of meaningful insights from raw data. Feature engineering and data preprocessing play a crucial role in improving the accuracy and interpretability of predictive models. Techniques such as transformation, normalization, and dimensionality reduction help identify the most relevant factors affecting crop productivity while reducing noise and redundancy in the data. In this study, feature engineering is used extensively to construct interaction variables, temporal trends, and transformed features, while Principal Component Analysis is explored to evaluate its impact on reducing feature dimensionality and improving model efficiency.

This study proposes a machine learning framework for crop yield prediction that integrates data preprocessing, exploratory analysis, feature engineering, and predictive modeling techniques. The proposed approach focuses on analyzing the relationships between climatic conditions and crop productivity while identifying the most influential factors affecting yield. Multiple machine learning algorithms are evaluated to determine their predictive performance, with particular emphasis on ensemble methods. The findings of this study are expected to provide valuable insights for improving agricultural decision-making and supporting sustainable food production systems in Bangladesh.

II. LITERATURE REVIEW

Crop yield prediction has become an important research area because of its role in improving agricultural planning and ensuring food security. Traditional statistical methods have long been used for yield estimation, but they often struggle to model the complex relationships between environmental variables and crop productivity. In recent years, machine learning and deep learning techniques have shown strong potential for predicting crop yield from large agricultural datasets that combine climatic, soil, and management-related variables [2], [8].

Early research demonstrated that machine learning algorithms can effectively model nonlinear relationships between climate variables and crop yield. Studies using rainfall, temperature, soil properties, and cultivation-area information showed that machine learning methods often outperform conventional regression approaches by capturing complex interactions among multiple environmental factors [6], [7]. These findings highlighted the importance of climate-aware predictive systems, especially in regions where agricultural output is highly sensitive to environmental changes [4], [11].

As research progressed, attention shifted toward comparing different machine learning algorithms for crop yield prediction. Comparative studies explored support vector machines, decision trees, random forests, and boosting-based ensemble methods, showing that ensemble learning approaches often achieve higher predictive accuracy because they combine the strengths of multiple models [4], [9], [10]. In particular, decision tree-based methods were found to be useful because of their interpretability and ability to handle heterogeneous agricultural datasets [14].

Beyond climatic factors, several studies emphasized the importance of soil characteristics and nutrient composition in determining crop productivity. Soil properties such as nitrogen content, potassium levels, and organic matter significantly affect plant growth, and recent research has incorporated such variables into predictive models to improve accuracy [5], [15]. In addition, explainable artificial intelligence methods and feature-importance analysis have been used to better understand how environmental and soil variables influence predictions, making model outputs more actionable for agricultural decision-making [5], [14].

Another important direction in the literature concerns feature engineering and data preprocessing. Agricultural datasets often include many variables gathered from different sources, which can introduce redundancy and noise. To address this issue, researchers have used dimensionality reduction and feature selection methods, including Principal Component Analysis, to identify the most relevant predictors and improve interpretability [6], [8]. Hybrid analytical frameworks that combine statistical analysis with machine learning have also been explored to improve predictive performance and robustness [12], [13].

With the growing availability of satellite imagery and remote sensing data, recent studies increasingly incorporate

geospatial information into crop yield prediction. Remote sensing allows large-scale monitoring of vegetation conditions, land characteristics, and crop health, which can substantially improve yield estimation across wide geographic regions [3], [18], [20]. Deep learning architectures are particularly well suited to such high-dimensional data and have shown strong performance in extracting complex patterns from climate and satellite inputs [2], [17], [19].

Recent work has also focused on integrating multiple data sources within unified predictive frameworks. By combining climate data, soil information, historical yield records, and remote sensing observations, researchers have developed more comprehensive systems for analyzing agricultural productivity [3], [15], [16]. Time-series and trend-based approaches have additionally been used to study long-term and seasonal yield variations, demonstrating the value of combining predictive modeling with temporal analysis [12], [18].

Overall, the literature shows that machine learning and deep learning techniques can significantly improve crop yield prediction by modeling complex relationships among climate, soil, and agricultural factors [8], [19]. At the same time, challenges remain in terms of data quality, model interpretability, computational cost, and integration of diverse datasets. Building on these studies, the present work develops a machine learning-based framework for Bangladesh that combines climate and agricultural variables, feature engineering, and comparative model evaluation to provide meaningful insights for agricultural planning and food-security support.

III. METHODOLOGY

A. Data Collection

The dataset used in this study was obtained from a publicly available Kaggle source, namely the Crop Yield Prediction Dataset. It contains historical agricultural records, including crop yield, climatic conditions, and agricultural inputs across multiple countries. For the purpose of this study, only the records corresponding to Bangladesh were selected. This region-specific filtering ensures that the analysis remains contextually relevant, given the importance of agriculture in Bangladesh's economy and food system.

B. Data Preprocessing

The preprocessing phase involved several steps to prepare the dataset for analysis and modeling. Initially, the dataset was filtered to retain only the records associated with Bangladesh. The column names were standardized for clarity and consistency, with key variables renamed as Yield, Rainfall, Pesticides, and Temperature.

Irrelevant columns, such as unique identifiers, were removed to eliminate redundancy. The dataset was examined for missing values and inconsistencies; however, no significant missing data was observed, allowing the study to proceed without imputation. Categorical variables, particularly the crop type (Item), were transformed using one-hot encoding. This conversion enabled the inclusion of categorical information in

numerical form, making it compatible with machine learning algorithms.

TABLE I
STATISTICAL SUMMARY OF DATASET VARIABLES

Statistic	Year	Yield (hg/ha)	Rainfall (mm/year)	Pesticides (tonnes)	Temperature (°C)
Count	294	294	294	294	294
Mean	2001.90	52,518.09	2666.00	6,609.50	26.10
Std Dev	7.12	46,650.72	0.00	5,294.83	0.28
Min	1990	7,339	2666.00	1,266.00	25.55
25%	1996	15,028	2666.00	1,919.00	25.91
50% (Median)	2002	29,358	2666.00	3,295.78	26.09
75%	2008	93,656	2666.00	12,936.50	26.28
Max	2013	193,790	2666.00	15,330.16	26.65

Table I presents the statistical summary of the dataset used in this study. The dataset consists of 294 observations spanning the years 1990 to 2013. Crop yield exhibits a wide range, with values varying significantly across different crops and years, as indicated by the high standard deviation. In contrast, rainfall remains constant throughout the dataset, suggesting limited variability for this feature. Temperature shows very low variation, indicating relatively stable climatic conditions over the observed period. Pesticide usage, however, demonstrates substantial variability, which may have a notable influence on crop yield. These observations provide important insights into the distribution and characteristics of the dataset, guiding subsequent feature engineering and modeling decisions.

C. Exploratory Data Analysis

Exploratory Data Analysis (EDA) was conducted to understand the distribution, variability, and relationships among the variables in the dataset. This step is essential for identifying patterns, detecting anomalies, and guiding feature engineering decisions.

A histogram of crop yield (Figure 1) was used to examine its distribution. The plot shows that yield values are widely spread, with a noticeable concentration in the lower and mid ranges and fewer observations at higher values. This indicates a slightly skewed distribution, suggesting that crop yield varies significantly across different crops and years. Understanding this distribution is important, as it provides insight into the variability of the target variable and helps determine whether transformations are needed for modeling.

To analyze temporal patterns, the yearly average yield was plotted (Figure 2). The line graph illustrates how crop yield changes over time, capturing both short-term fluctuations and long-term trends. While some variations can be observed across different years, the overall trend remains relatively stable, indicating that yield is influenced by multiple interacting factors rather than a simple linear progression over time. This observation supports the need for models capable of capturing complex relationships.

To further investigate relationships between variables, a correlation matrix was computed and visualized using a heatmap (Figure 3). The heatmap summarizes the pairwise linear relationships among yield, rainfall, temperature, and pesticide usage. From the visualization, it can be observed

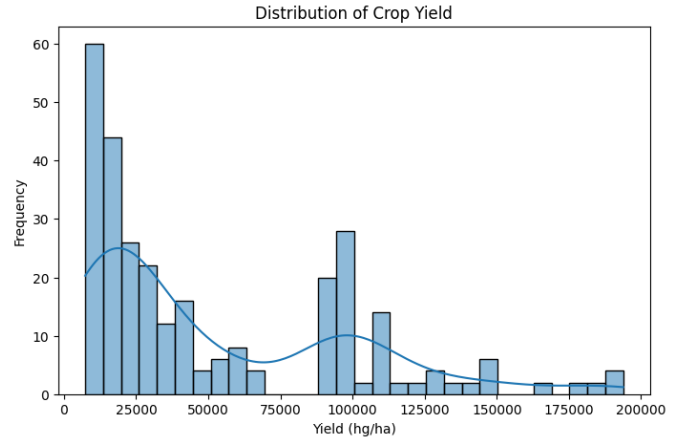


Fig. 1. Distribution of Crop Yield

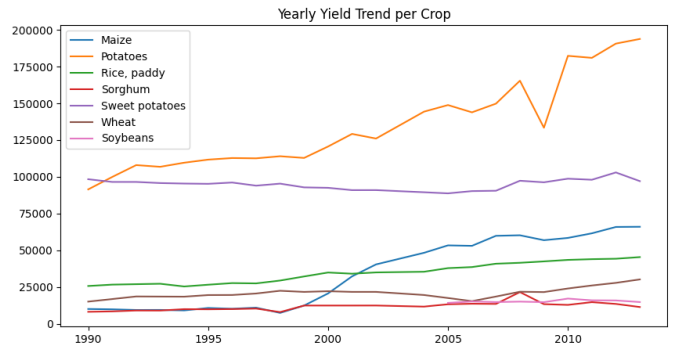


Fig. 2. Yearly Average Crop Yield Trend

that some variables exhibit weak to moderate correlations with yield, indicating that no single factor alone determines crop productivity. Additionally, rainfall appears to have little to no variation in the dataset, which limits its contribution as a predictive feature. This reinforces the importance of combining multiple variables and applying feature engineering techniques to capture meaningful patterns.

In addition to these visualizations, further exploratory analysis was performed using scatter plots and boxplots (not shown in this report). Scatter plots between yield and variables such as rainfall, temperature, and pesticide usage revealed non-linear relationships, suggesting that simple linear models may not be sufficient. Boxplots were used to detect outliers and assess variability across features, ensuring that extreme values do not adversely impact model performance.

Overall, the EDA process provided valuable insights into the structure and characteristics of the dataset. It highlighted the presence of variability in yield, limited variation in certain features, and complex relationships among variables. These findings informed the subsequent feature engineering and model selection steps.

D. Feature Engineering

To enhance the predictive capability of the models, several new features were engineered based on domain knowledge

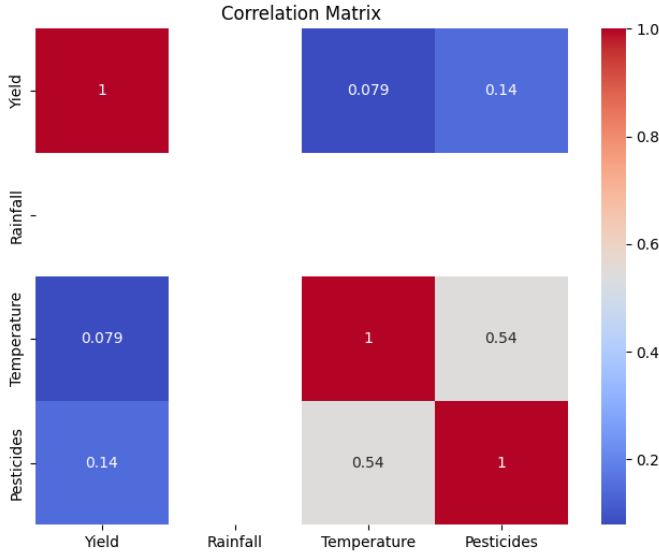


Fig. 3. Correlation Heatmap of Key Variables

and statistical reasoning. Interaction features were introduced to capture combined effects of climatic variables, such as the product of rainfall and temperature. A feature representing pesticide usage relative to rainfall was also created to account for environmental dependency.

Temporal features were incorporated to model trends over time, including a linear trend variable and a polynomial representation of the year. These features enabled the models to capture both gradual and non-linear temporal patterns in crop yield. The features were also normalized for models dependent on distance and sensitive to scaling.

Furthermore, logarithmic transformations were applied to rainfall and pesticide variables to reduce skewness and stabilize variance. This transformation improves model performance by making the data distribution more suitable for learning algorithms. Categorical encoding was performed using one-hot encoding on the crop variable, allowing the models to distinguish between different crop types effectively.

E. Dimensionality Reduction Using Principal Component Analysis

To further investigate the effect of feature dimensionality on model performance, Principal Component Analysis (PCA) was applied as an optional dimensionality reduction technique. PCA transforms the original set of possibly correlated features into a smaller number of uncorrelated components while preserving most of the variance present in the data.

Prior to applying PCA, the feature set was standardized to ensure that all variables contributed equally to the transformation. The number of principal components was selected based on a cumulative explained variance threshold of 95%, ensuring that the majority of the original information was retained.

The primary objective of incorporating PCA was to evaluate its impact on models sensitive to high-dimensional data, such

as K-Nearest Neighbors and Support Vector Machine. However, tree-based models like Random Forest were trained on the original feature space, as they inherently handle feature selection and are resilient to multicollinearity.

F. Model Development

The dataset was divided into training and testing subsets using an 80:20 split to evaluate model generalization. Feature scaling was applied using standardization for models that rely on distance calculations, specifically K-Nearest Neighbors (KNN) and Support Vector Machine (SVM).

Four supervised learning algorithms were implemented:

- Random Forest Regressor
- AdaBoost Regressor
- K-Nearest Neighbors Regressor
- Support Vector Machine Regressor

The Random Forest model was configured with 150 estimators and a maximum depth of 15, enabling it to capture complex relationships while controlling overfitting. AdaBoost utilized decision trees as base learners, focusing on sequential error correction. KNN was implemented with seven neighbors, while SVM employed a radial basis function kernel with tuned hyperparameters.

G. Model Evaluation

Model performance was evaluated using Root Mean Squared Error (RMSE) and the coefficient of determination (R^2 score). These metrics provide a comprehensive assessment of prediction accuracy and explanatory power.

To compare model performance, results were summarized in tabular form and visualized using bar charts. Additional diagnostic plots, including actual versus predicted values and residual plots, were used to assess model behavior and identify potential biases. Feature importance analysis was conducted for the Random Forest model to identify the most influential predictors. This analysis provided insight into the relative contribution of climatic and engineered features in determining crop yield.

H. Model Validation

To assess the robustness and generalization capability of the best-performing model, 5-fold cross-validation was applied to the Random Forest model. This technique evaluates the model across multiple data splits, providing a more reliable estimate of performance.

In addition, a learning curve was generated to analyze how model performance varies with training data size. This helped identify whether the model suffers from overfitting or underfitting and whether additional data could further improve performance.

IV. RESULTS AND DISCUSSION

A. Model Performance Evaluation

The performance of the implemented machine learning models was evaluated using Root Mean Squared Error (RMSE) and the coefficient of determination (R^2 score). These

metrics provide complementary insights into model accuracy and explanatory power.

TABLE II
MODEL PERFORMANCE COMPARISON

Model	RMSE	R^2 Score
Random Forest	933.32	0.9995
AdaBoost	3154.45	0.9948
KNN	3979.25	0.9918
SVM	45092.83	-0.0588

The results indicate that ensemble-based models outperform other approaches, with Random Forest achieving the highest predictive accuracy.

B. Comparative Analysis of Models

A comparative evaluation of the models highlights clear differences in their ability to capture complex relationships within the dataset. The Random Forest model achieved the best performance, with the lowest RMSE and highest R^2 score, indicating an excellent fit to the data. This can be attributed to its ensemble structure, which allows it to model non-linear relationships and interactions among features effectively. AdaBoost also demonstrated strong performance, though with slightly higher prediction error, suggesting comparatively lower robustness.

The K-Nearest Neighbors model produced reasonably high accuracy but exhibited higher error values, reflecting its sensitivity to feature scaling and local data structure. In contrast, the Support Vector Machine model performed poorly, with a negative R^2 score, indicating its inability to capture the underlying patterns in the dataset. This is likely due to the complexity of the data and the limitations of SVM in handling high-dimensional feature spaces without extensive hyperparameter tuning.

Overall, the results confirm that ensemble learning methods are better suited for this problem, as they can effectively handle feature interactions and non-linear dependencies present in agricultural and climatic data.

C. Cross-Validation Results

To assess the robustness of the best-performing model, 5-fold cross-validation was conducted on the Random Forest model. The results are presented in Table III.

TABLE III
CROSS-VALIDATION PERFORMANCE OF RANDOM FOREST

Fold	R^2 Score
1	0.9859
2	0.9932
3	0.9290
4	0.9481
5	0.9160
Mean	0.9544

The mean cross-validation score of 0.9544 indicates strong generalization capability. Although slightly lower than the test R^2 score, the consistency across folds suggests that the model is stable and not excessively overfitted.

D. Feature Importance and Model Behavior

Feature importance analysis from the Random Forest model revealed that climatic variables such as rainfall and temperature are among the most influential predictors of crop yield. Additionally, engineered features, including interaction terms and logarithmic transformations, contributed significantly to model performance. This highlights the importance of feature engineering in capturing complex relationships within the data.

E. Learning Curve Analysis

The learning curve of the Random Forest model 4 provides insight into its training behavior as the size of the training data increases. The training score remains consistently high, while the validation score improves steadily with increasing data. This suggests that while additional data may further improve performance, the model has already captured the essential patterns within the dataset.

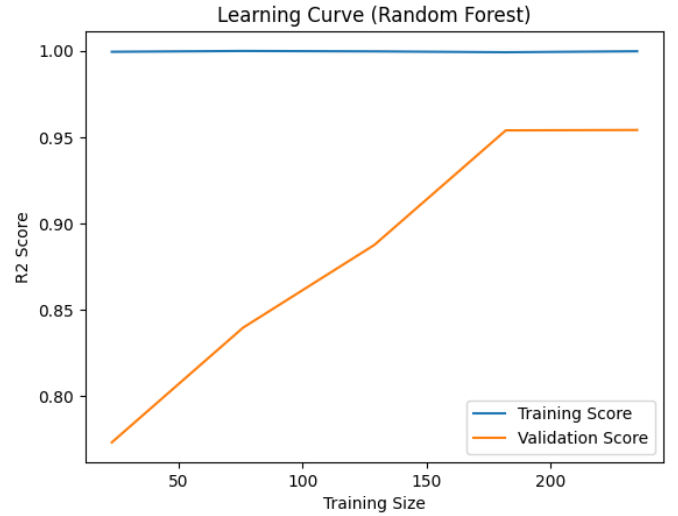


Fig. 4. Distribution of Crop Yield

F. Key Insight

The findings suggest that ensemble learning methods, combined with effective feature engineering, provide a highly reliable approach for crop yield prediction, while dimensionality reduction techniques such as PCA offer limited benefit when the feature set is already well-engineered.

V. CONCLUSION

This study presented a machine learning-based framework for predicting crop yield using agricultural and climatic data specific to Bangladesh. The approach combined data preprocessing, exploratory data analysis, and feature engineering to construct a robust predictive pipeline. Several supervised learning models, including Random Forest, AdaBoost, K-Nearest Neighbors (KNN), and Support Vector Machine (SVM), were implemented and evaluated using standard regression metrics.

The experimental results demonstrated that ensemble learning methods, particularly Random Forest, significantly outperform other models in terms of prediction accuracy. The Random Forest model achieved the highest R^2 score and the lowest RMSE, indicating its strong ability to capture complex, non-linear relationships within the dataset. In contrast, SVM failed to generalize effectively, highlighting its limitations in handling this type of data without extensive tuning.

Feature engineering played a crucial role in enhancing model performance. The introduction of interaction features, temporal variables, and logarithmic transformations allowed the models to better capture underlying patterns in the data.

Overall, the findings confirm that combining domain-informed feature engineering with ensemble learning techniques provides an effective solution for crop yield prediction. The framework developed in this study can serve as a valuable tool for understanding the impact of environmental factors on agricultural productivity.

VI. FUTURE WORK

While the current study demonstrates strong predictive performance, several directions for future improvement can be identified.

One key limitation of the present work is the dependency on a single primary dataset with a relatively fixed set of features. In future work, the models can be trained on a more refined and enriched dataset that incorporates additional variables. Integrating higher-quality and more granular data is expected to improve model accuracy and generalization.

Furthermore, the dataset used in this study primarily reflects aggregated historical data. Future research could explore the use of real-time or high-resolution data sources to capture dynamic environmental conditions more effectively. Another potential extension is the incorporation of time-series forecasting methods to complement the current feature-based models.

Although PCA was explored as a dimensionality reduction technique, alternative feature selection methods or hybrid approaches could be investigated to further optimize model performance.

Finally, the proposed framework can be extended to region-specific or crop-specific settings to evaluate its scalability and adaptability. Such extensions would enhance its applicability as a generalized decision support tool for agricultural planning and policy development.

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